

CALENDARING QUALITY IMPROVEMENT AT RUBBERCRAFT

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Abstract

Having high variations of manufactured products affects the quality of the product. And with no way to monitor this variation, no improvements will be made for the most part. This is what is happening in Rubbercraft within their rubber calendaring process. So, creating one was necessary to be able to monitor this variation. With the use of individual moving-range control charts, the quality improvement senior project team was able to monitor any variation that is out-of-control. To optimize the process of data collecting, the team proposed the use of data logging technology to automatically send measurements to a database to be stored and created into control charts. After coming up with this process of developing control charts, a control measure had to be created and put in place. So, an out-of-control action plan was created to create standard operating procedures once an out-of-control data point was discovered on the control chart. This whole process granted valuable information on the rubber calendaring process, because it showed whether improvements can be made or not, and that a plan can be made to implement these improvements.

Acknowledgments

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Table of Contents

Table of Contents.....	6
1.0 Introduction.....	10
2.0 Body.....	11
2.1. Approach.....	11
2.1.1. Define	11
2.1.2. Measure	11
2.1.3. Analyze	11
2.1.4. Improve.....	12
2.1.5. Control.....	12
2.2. The Challenge: Uncontrolled Variation	12
2.3. Sampling Data & Initial Analysis	13
2.3.1. Control Charts.....	13
2.3.2. Capability Analysis.....	17
2.4. Alternative 1: Mitutoyo Data Transfer Device.....	18
2.4.1. Implementation Overview	18
2.4.2. Technical Advantages	19
2.4.3. Practical Considerations	19
2.5. Alternative 2: Arduino Measurement Device.....	20
2.5.1. Implementation Overview	20
2.5.2. Technical Advantages	22
2.5.3. Limitations	22
2.6. Database Construction and Management	23
2.6.1. Technical Constraint.....	27
2.7. Economic Analysis	27
2.8. Monitoring and Process Control	32
2.9. Out-of-Control Action Plan	34
2.10. Additional Recommendations	37
2.10.1. Predictive Maintenance Program	37
2.10.2. Workforce Development	37
2.10.3. System Enhancements	38
2.10.4. Continuous Improvement Process.....	38
2.11. Standards and Constraints.....	39
3.0 Reflective Writing.....	41
3.1. Introduction.....	41
3.2. Body.....	42
3.3. Conclusion	43
4.0 Conclusion and Recommendations	44
5.0 References	46

6.0	<i>Appendix</i>	<i>47</i>
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List of Tables

Table 1 – Process Capabilities of the Revised Individual Moving-Range Chart	18
Table 2 – Components List for Arduino Sensor Device.....	21
Table 3 – Alternative 1 Cost Components.....	28
Table 4 – Alternative 2 Cost Components.....	28
Table 5 – Savings Components.....	29
Table 6 – Net Present Value of Alternative 1	30
Table 7 – Net Present Value of Alternative 2	30
Table 8 – Internal Rate of Return Cashflows.....	31

List of Figures

Figure 1 – Histogram of Variance CTQ	14
Figure 2 – Probability Plot of Collected Data.....	14
Figure 3 – Original Individual Moving-Range Chart	15
Figure 4 – Revision of Individual Moving-Range Chart.....	16
Figure 5 – Process Capability Report of the Revised Version of Individual Moving-Range Chart.....	17
Figure 6 – Mitutoyo Gauge on Left, Mitutoyo U-Wave on the right	19
Figure 7 – Diagram of the Arduino Sensor Device	20
Figure 8 – 3D CAD Model of Mounting Fixture for Arduino Sensor Device	21
Figure 9 – Prototype of Arduino Sensor Device & Close-up of Wiring Scheme.....	23
Figure 10 - Entity Relationship Diagram for the Structure of the Database.....	25
Figure 11 - Material Thickness Data Entry Form.....	26
Figure 12 - Query Based on the Compound Number	27
Figure 13 – Out of Control Action Plan.....	35
Figure 14 – Swim Lane Diagram, Halved for Viewability; Full Diagram in End Section of Report	36
Figure 15 – Full View of the Swim Lane Diagram	47
Figure 16 Data Entry Part Number Form	47
Figure 17 Data Entry Order Form.....	47
Figure 18 Data Entry Manufacturing Order Form.....	48
Figure 19 Data Relationships.....	48
Figure 20 Data Entry Customer Form	49
Figure 21 Material Thickness Algorithm Section 1.....	Error! Bookmark not defined.
Figure 22 Material Thickness Algorithm Section 2.....	49

1.0 Introduction

The company Rubbercraft makes a high variety of rubber components, each having their own specifications and required calendared thickness. However, in this manufacturing process, variations are bound to happen and when they do occur, the company does not do anything to monitor this process. Doing so can ensure that quality is improved, variation is minimized, and improvements can be made. The issue pertains to the quality of material that exits the calendaring process at Rubbercraft. Known defects to exist within the calendaring process include voids (holes), impressions, thickness inconsistencies, and air bubbles in the materials. This project aimed to study the Critical to Quality Characteristic (CTQ), material thickness. Rubbercraft is a rubber manufacturer with job shop style facility. They specialize and services industries like aerospace, defense, energy, and power. The company has a vast product line. To name a few products, Rubbercraft produces complex seals, gaskets, and O-rings. Engineering characteristics and tolerance are specified for the particular application of the product. Parts are designed in this manner. It is of utmost importance when parts are manufactured, they meet the specified requirements. When products fail to meet requirements, parts are scrapped and remade. Calendaring at this facility involves feeding a compound that's approximately the size and shape of a basketball into rollers set to a specific gap. Under high pressure, the malleable material is flattened, producing sheets of rubber. (Wadhwa, 2021). A sheet of material is stored as a roll. Later the rolled sheets are laid out and cut into strips or pieces, which are then placed into molds to produce parts. These sheets of rubber material must maintain precise thicknesses to ensure

consistency and quality in the final product. No system previously existed to record material thickness and continuously monitor the process over time.

2.0 Body

2.1. Approach

The approach to this particular project utilizes the IISE DMAIC Methodology.

Each letter of DMAIC represents an individual project objective that will contribute to completion and success when tackling problems in manufacturing.

2.1.1. Define

The first objective is to “Define” the problem. In this case, the problem being approached at Rubbercraft is an observed high frequency of defects and scrapped manufactured goods related to rubber being released from their calendaring operation. Defining the issue at hand properly will set work towards an effective solution into motion.

2.1.2. Measure

The next objective is to “Measure” an output from this operation. The output being measured from the calendaring process is material thickness. Samples of compressed rubber material will be measured using a company calibrated thickness gauge. This thickness data will then be documented for comparison with the target thickness specified by the manufacturer.

2.1.3. Analyze

After the sample data is documented, this data will be analyzed using a variety of tools in the “Analyze” objective. This project will utilize the use of individual and moving range control charts that display the data in

relation to control limits that will indicate whether the process is in control or out of control. Capability analyses will then be conducted to determine if the process effectively meets the level required by the customer. The usage of these tools will provide valuable information essential to creating solutions that will combat the initial problem.

2.1.4. Improve

The “Improve” objective involves using the data and conclusions gathered from the “Analyze” phase to draft solution alternatives that Rubbercraft can feasibly integrate into their manufacturing operations. These solutions are meant to be data-driven, comply to current company operational constraints, and effectively mitigate the issues defined in their daily operations.

2.1.5. Control

The final objective is “Control”, which is meant to monitor the solutions once integrated into their operation to ensure that they are achieving their intended outcomes and functionality.

2.2. The Challenge: Uncontrolled Variation

Rubbercraft had issues within the company that affected the quality of their production. There was a lot of variation in the difference between actual part thickness and their target thickness within the calendaring process. This was due to the varying skills of the different laborers that worked there. Quality was then affected, and reworks had to be done more often. The challenge of the project was that the company did not have any process of monitoring or tracing their defects within the rubber calendaring system. Thus, defects would go unnoticed,

and quality would remain unchecked. The team created a system for the company that would allow its operations to track the quality defects through individual & moving range charts, along with an out-of-control action plan, to counteract this lack of awareness within the process.

2.3. Sampling Data & Initial Analysis

The sampling efforts conducted by the team over a period of two months yielded 43 usable datapoints. These datapoints represent material thickness measurements acquired using a thickness gage on the calendaring manufacturing floor. Actual thicknesses, target thicknesses specified by the manufacturer, manufacturing dates, and manufacturing order numbers were obtained during the initial data collection phase. The team was then able to construct individual and moving range control charts using the datapoints collected. The individual charts visualize variation between the target and actual thickness, display calculated control limits which are indicators of process performance. It was discovered that noticeable variation was present in the operation, and two datapoints were determined to be out of control.

2.3.1. Control Charts

To monitor the variation within the company's rubber calendaring process, a control chart must be utilized to ensure stability of the process. This would allow whoever oversees quality control within the company to monitor and identify out-of-control points within the chart. It was necessary to first ensure that the distribution of the collected data follows normal distribution. So, using a histogram seen in Figure 1, the team was able to observe that the

data was in normal distribution. This was proven by the probability plot that was created to match the data of the histogram, because the data followed a trend of a normal distribution at a 95% confidence interval, as shown in Figure 2.

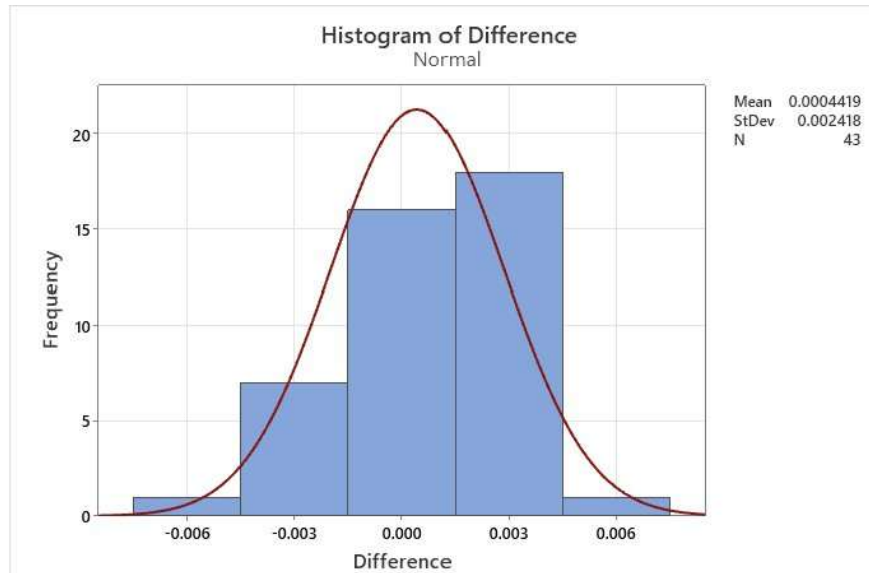


Figure 1 – Histogram of Variance CTQ

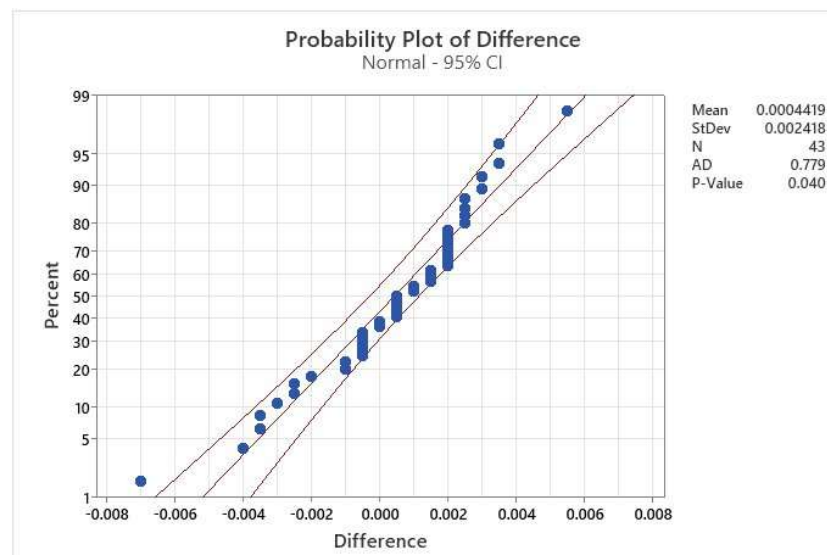


Figure 2 – Probability Plot of Collected Data

The chart that the team decided to use was the individual & moving range charts (also known as I-MR charts) to do so. This was because the method of sampling used only collected one point within each of the different calendared rubber sheets. Other charts could be used, such as x-bar charts, but since the process of collecting samples was tedious and time-consuming with limited amounts of time, the team decided that the I-MR chart would be the best way to move forward with the project. In the development of this chart, the team came up with two different charts: one summarizing all the collected data, and the other that had revisions and omitted the out-of-control points within the system. The chart that included all the data is shown in Figure 3 and the chart that made revisions is shown in Figure 4.

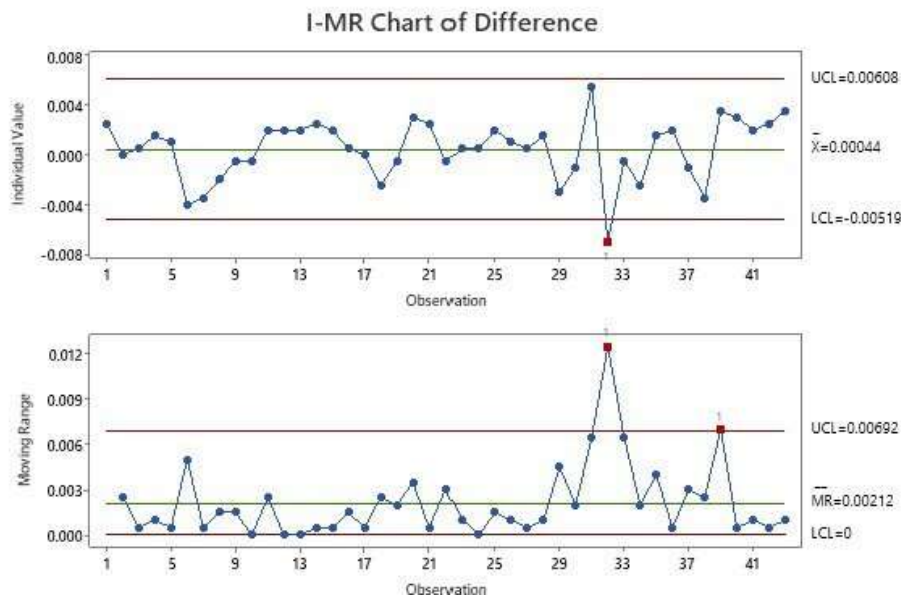


Figure 3 – Original Individual Moving-Range Chart

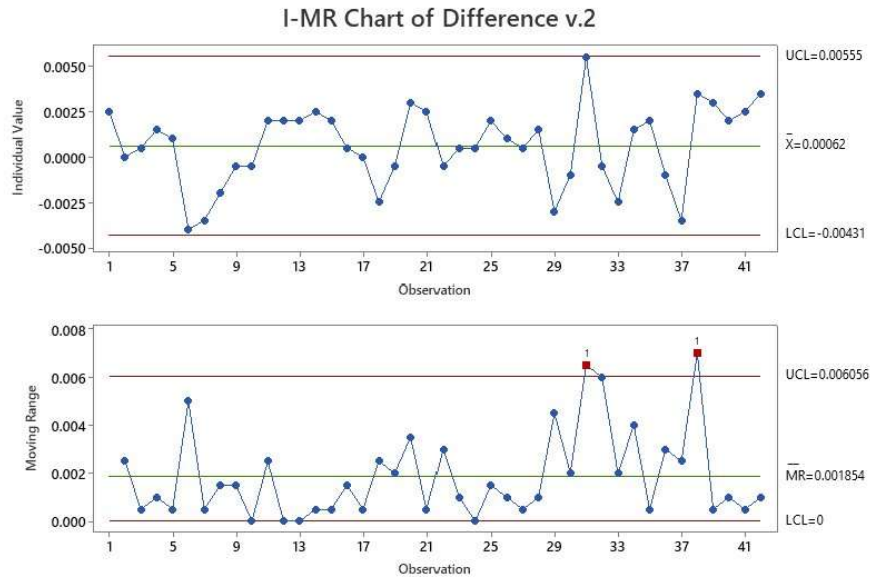
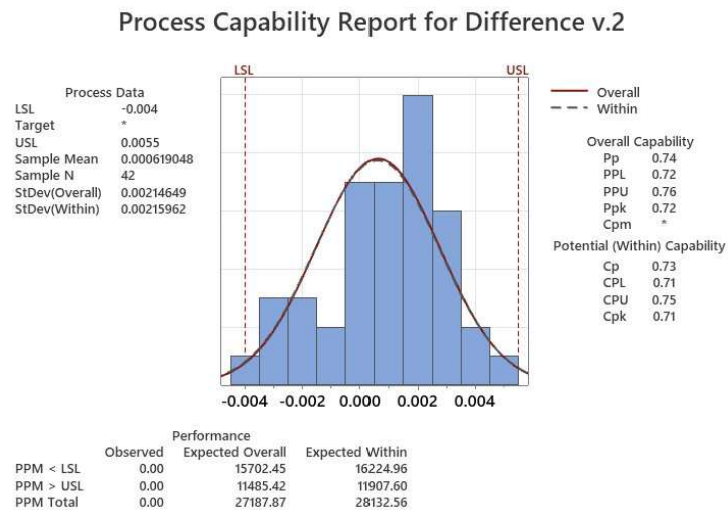


Figure 4 – Revision of Individual Moving-Range Chart

In the original I-MR chart, one can observe that there are three total out-of-control points within the collected data. The I-MR chart was revised into the Figure 4 chart and this was based on assumed assignable causes. This was due to there being no form of tracing available to the out-of-control data points within Rubbercraft. In other words, the company had no form of tracking what went wrong with the defects. So, once the chart was revised, it showed no out-of-control points in the individual chart, which is great. Also, the moving range chart had two out-of-control points. But this is acceptable because the company works in a job shop layout. In other words, the company has a high variety of products being made and a low-to-medium quantity of parts manufactured. The results given in this control chart is stable, even though the moving range chart has a couple points that are out of control. This is because of knowing that the process is a job shop layout, it can be assumed that these points are in high variability for good reason.

2.3.2. Capability Analysis

In the next part of analysis, the team wanted to see how well the process of calendaring was, and specifically, if any improvements can be made. To measure this performance, process capability reports were created to do so. Process capability reports are used in manufacturing and quality control, when a person wants to see if the parts manufactured are within specification limits. First, the team had to make sure the control chart was stable. Looking at the out-of-control points and the trends of both the individual and moving range charts, it can be assumed that the process is stable. Because of this, the team went ahead and created a process capability report, one detailing the original IMR chart and the other that went over the revised IMR chart, which was more accurate. In the revised process capability report, one can see that in Figure 5 there is data that is out of specification limits in the histogram.



The actual process spread is represented by 6 sigma.

Figure 5 – Process Capability Report of the Revised Version of Individual Moving-Range Chart

Moreover, the process capability signals that the process is not capable of producing rubber sheets within specification limits. This shows that improvements can be made. Additionally, the defect rate is at 2.72% for parts out of specification limits. This is the defect rate within the calendaring process. This is all valuable information for the company to plan out the next steps to take when making improvements to the calendaring process. Because the revised report was used instead of the original report, more accurate data is given. So, there is no need to go in depth on the original report.

Table 1 – Process Capabilities of the Revised Individual Moving-Range Chart

Process Capability	Rate
C_p	0.73
C_{pl}	0.71
C_{pu}	0.75
C_{pk}	0.71

2.4. Alternative 1: Mitutoyo Data Transfer Device

The issue we are trying to resolve is uncontrolled variation. To improve the accuracy and traceability of thickness measurements in the calendaring process, the first proposed solution involves integrating a Mitutoyo data transfer device with the existing Mitutoyo thickness gauge. This device enables automated data logging, reducing human error and streamlining quality control.

2.4.1. Implementation Overview

Our first proposed solution enhances measurement accuracy through a Mitutoyo data transfer system that interfaces with Rubbercraft's existing thickness gauge. This system consists of three key components:

- Mitutoyo Data Logger – Logs thickness measurements from manual gauge
- Automated Database Entry – Eliminates manual transcription errors
- Visual Monitoring Display – 40" HD screen provides real-time feedback

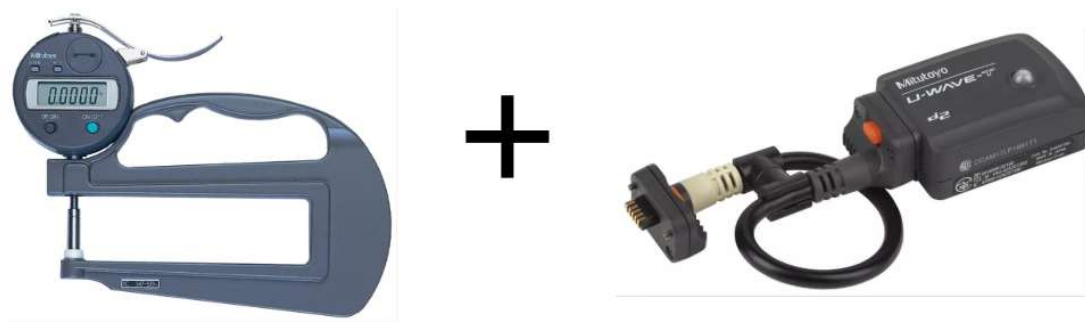


Figure 6 – Mitutoyo Gauge on Left, Mitutoyo U-Wave on the right

2.4.2. Technical Advantages

- Seamless Compatibility: Retrofits existing equipment
- Error Reduction: Removes human recording inaccuracies
- Historical Tracking: Creates searchable quality records for trend analysis

2.4.3. Practical Considerations

- Implementation Ease: Can be operational within one production shift
- Training Requirements: Less than 30 minutes for operators
- Cost Efficiency: Significant ROI through scrap reduction

This alternative is simpler & more practical. It does not reinvent the process, simply makes it more efficient.

2.5. Alternative 2: Arduino Measurement Device

For a more advanced approach, the team proposes a custom Arduino-based measurement system using VL53L0X time-of-flight (ToF) sensors. This system automates thickness calculations without direct operator intervention, improving precision and repeatability. In terms of how it works, the rubber sheet passes between the two VL53L0X sensors and the thickness is calculated within the Arduino. The process follows the diagram seen below, in Figure 7.

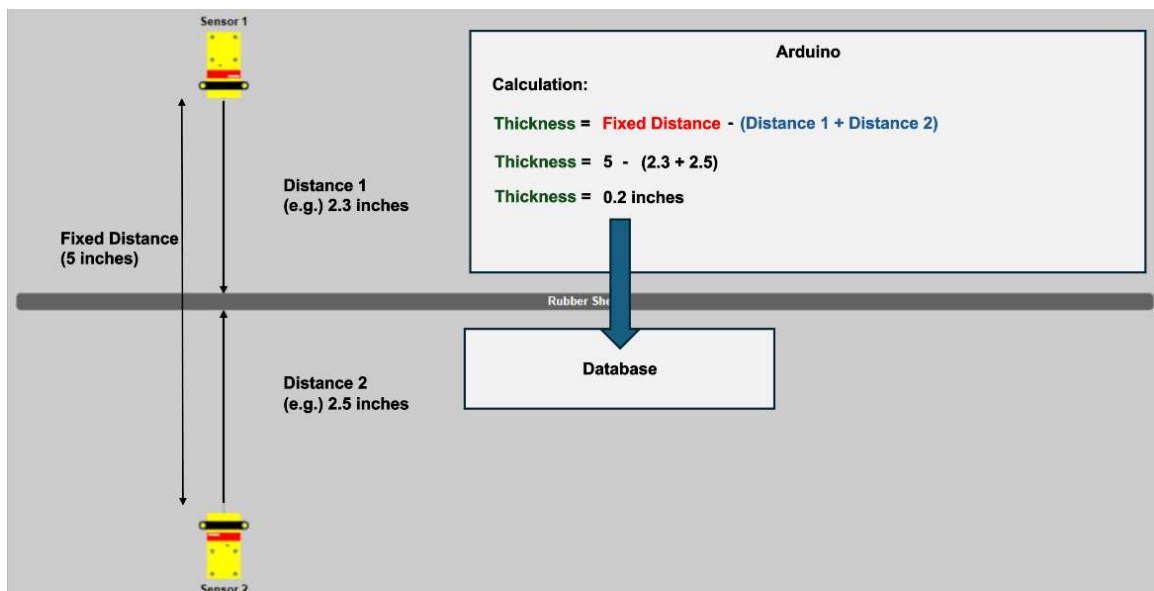


Figure 7 – Diagram of the Arduino Sensor Device

2.5.1. Implementation Overview

Moving on to what is required to build such a system, we have a list of the components in Table 2 with their associated costs. Furthermore, a computer and an Arduino IDE are required to make this solution possible.

Table 2 – Components List for Arduino Sensor Device

Material	Quantity Needed	TOTAL
VL53L0x	2	\$29.90
Arduino Uno Rev 3	1	\$6.22
830 Point Breadboard	1	\$2.49
MtM Ribbon Cable	12	\$0.58
MtF Ribbon Cable	10	\$0.58
		\$39.77

Key components:

- VL53L0X Sensors: High Precision laser-ranging sensors for non-contact
- Arduino Microcontroller: Processes sensor data and calculates thickness
- Data Logging & Display: Measurements are sent to a database and displayed on a 40” HD monitor for operator feedback
- Mounting Fixture: Holds Arduino sensor device and connects to calendar

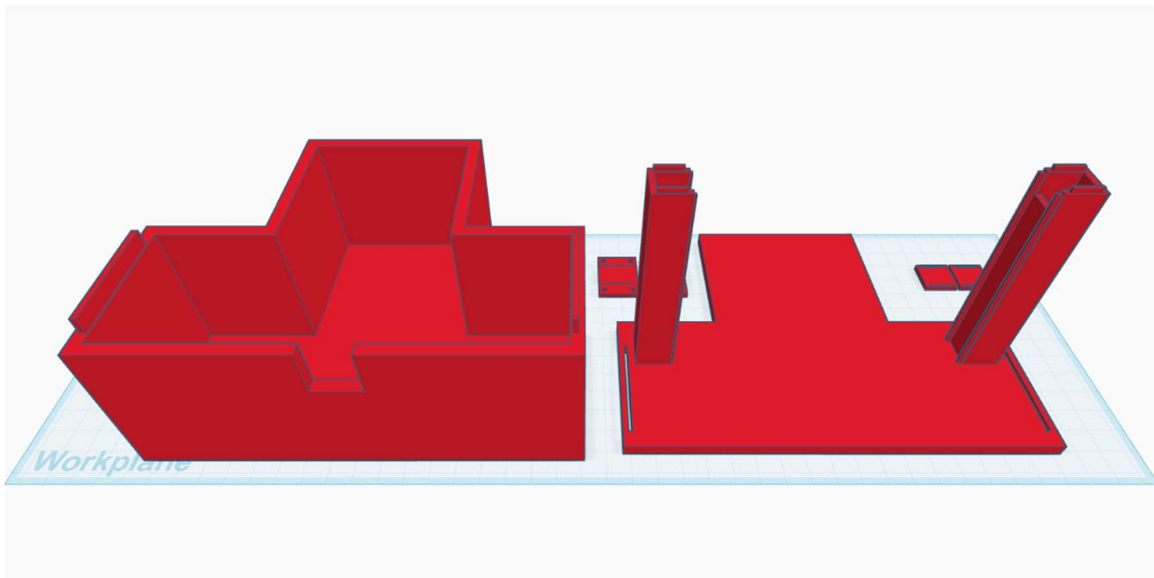


Figure 8 – 3D CAD Model of Mounting Fixture for Arduino Sensor Device

2.5.2. Technical Advantages

- Fully Automated Measurement: Eliminates human measuring error
- Higher Precision: Provides accurate readings without pressing material
- Scalability: Adapts for different rubber sheet widths and thicknesses

2.5.3. Limitations

- Development Complexity: Requires calibration and testing
- Integration Effort: Must be incorporated into the existing workflow
- Technical: Requires soldering, wiring, and programming knowledge

This solution offers a higher degree of automation but requires more initial setup compared to the Mitutoyo-based approach.

2.5.4. Program

Base code can be read in Figure 21 and Figure 22. The base program is from an Arduino forum by a user named “4nodews”, credit goes to them for the . Original program reads out both sensors and returns the values individually. Our team modified this program to meet the needs of our project.

- Variables:
 - Setdistance -
- Libraries:
 - Wire.h – Allows usage of multiple devices (two sensors)
 - VL53L0X.h – Reusable prewritten code for sensor functionality
- Modifications: Displays “Thickness” by taking both sensor readouts and subtracting them from a set distance. Added tare button functionality.

- Optional: Tare button can be made from any switch (cost is negligible and passed to the ribbon cables) but requires a pull-up resistor. Button in Figure 9 is not made with a pull-up resistor.

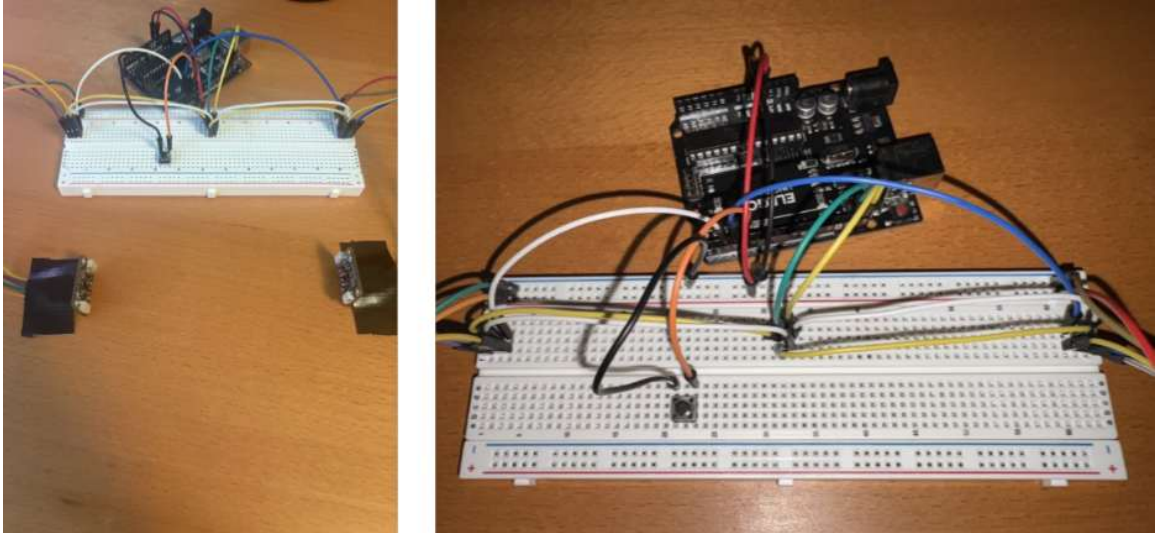


Figure 9 – Prototype of Arduino Sensor Device & Close-up of Wiring Scheme

2.6. Database Construction and Management

Material thickness is a potential key contributor to out of tolerance dimensions in part production. The purpose of the database is to store material thickness measurements for production control. The method of continually monitoring the thickness must be established. Utilizing Microsoft Access to design and implement the system's database structure. The database should continuously monitor the material thickness, securely store data, provide an intuitive user interface, and support efficient development of control charts. Control charts generated from data stored in the database help identify possible sources of variation in the operation that can be improved. The intended user for data entry into the database are the floor operators. The structure of the database contains

five tables. The tables represent entities. Entities are objects like people, events, or things that are meant to be described within the table from the types of information that is stored. Attributes are the columns in the table that help define characteristics of that particular entity. Entities are linked to one another with many-to-many or one-to-many relationships noted with the single and triple branch connects between the tables. Single refers to one, while triple represents many. Relationships between entities are established through the use of primary and foreign keys. The entity relationship diagram shown in Figure 10, contains information to describe customers, orders, parts, manufacturing orders, and measurements. Primary and foreign keys are noted in color. To note a few; customer name, customer address, part number, part description, and part order quantity. It is important to note that a specific order of data entry is required. The correct sequence is Customer Information, Part Details, Order Information, Manufacturing Order Information, and finally, Measurement Data, which can then be collected and recorded.

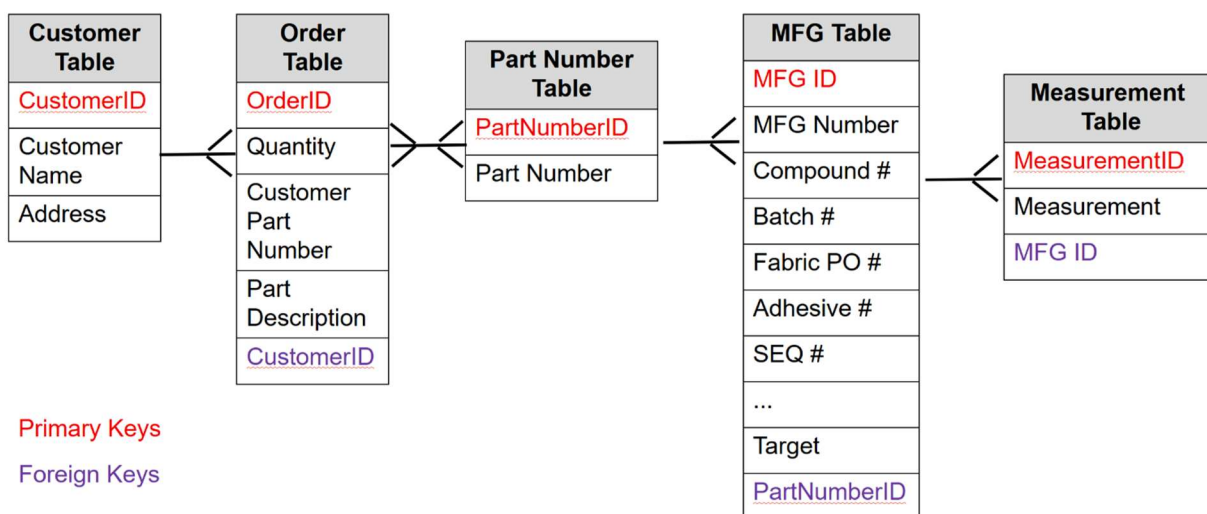


Figure 10 - Entity Relationship Diagram for the Structure of the Database

With the use of a form, thickness measurement data can be transferred immediately from the measurement site into the software as soon as it is recorded. Figure 11 references the interface of the form. The operator can use the dropdown menu to select the manufacturing order to be processed. Each MFG number is a unique six-digit identifier. Additionally, the timestamp of each measurement is automatically recorded when data is entered into the field. Depending on the length of the sheet of material, measurement points are to be captured at one-foot intervals.

Compound Query X MFGform X MEASUREMENTform X

ThicknessMeasurement

MEASUREMENTID 11

MFG# 112001

Measurement

- 0.0371
- 0.0351
- 0.0349
- 0.0355
- 0.0334
- 0.0344
- 0.0340
- 0.0329
- 0.0340
- 0.0390
- 0.0333
- 0.0370
- 0.0339
- 0.0349
- 0.0350
- 0.0390
- 0.0309
- 0.0399
- 0.0345
- 0.0355

CLOCK IN 4/17/2025 1:09:35 AM

Submit Clear

Record: 1 of 1 Search

Figure 11 - Material Thickness Data Entry Form

Using a query, the system retrieves the necessary data types required to generate control charts in order to monitor the process effectively. Figure 12 displays a query based on a specific compound number. Important to note queries for data based on specific compound, date, or specific thickness can provide valuable insights when analyzed to understand the behavior of the process. For example, a trend of out-of-tolerance points may appear at a specific point in time, for a

particular compound, or at a specific thickness, and the data allows this trend to be identified and analyzed.

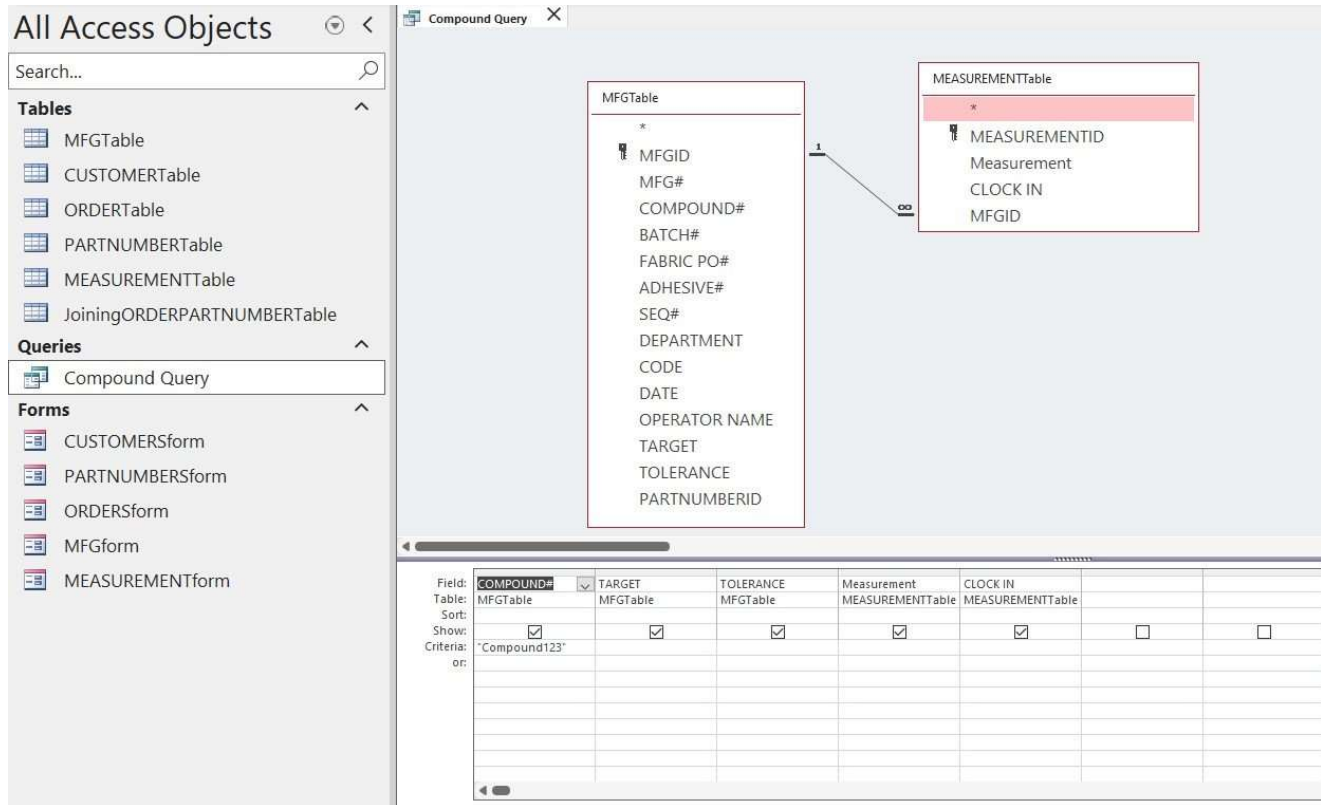


Figure 12 - Query Based on the Compound Number

2.6.1. Technical Constraint

One constraint with the database is a deeper understanding of the system built in the company is required to successfully integrate the proposed database for streamline and effective data communication within systems.

2.7. Economic Analysis

The economic analysis utilized for this project is a comparison of net present values and internal rates of return modeled over a one-year span of solution

integration. This required the modeling of cost and savings components. The cost components represent the initial investment made by the company in purchasing the solution, whereas the savings components represent the addition to the company's current revenue from the reduction in scrap parts the solutions posed will yield.

Table 3 – Alternative 1 Cost Components

Alternative 1	
Mitutoyo Data Transfer Device	\$400.00
40" HD Display with Stand	\$204.49
Engineering Testing and Development	\$50,000.00
Total	\$50,604.49

Table 4 – Alternative 2 Cost Components

Alternative 2	
Arduino Measurement Device	\$39.77
40" HD Display with Stand	\$204.49
Engineering Testing and Development	\$50,000.00
Total	\$50,244.26

The construction of initial cost components began with acquiring hardware costs for both alternative solutions. Alternative 1 is defined as the data logging and transfer device implementation, whereas alternative 2 is defined as the modification to the manufacturing point of measurement through the Arduino

based sensing device. Standard costs that will be required in both alternatives include engineering testing and development cost which will be done by Rubbercraft, and a 40” HD display that will illustrate the real-time status of the operation on the manufacturing floor.

Table 5 – Savings Components

Savings Components	
Dollar Value of Scrap Part 1	\$198.60
Dollar Value of Scrap Part 2	\$102.67
Average Dollar Value	\$150.64
Total Monthly Production from Company Data	55,104 parts
Calendar Scrap Rate	2.72%
Monthly Value of Parts Shipped	\$8,300,591.04
Total Monthly Benefit Yielded from Solution	\$225,776.08

A monthly benefit expected to be yielded from the implementation of either solution must be calculated through the formation of savings components. These savings components are a combination of data acquired directly from the company and a scrap rate obtained through the normal operating condition illustrated by the Capability Analysis. The total savings benefit per month was estimated to be \$225,776.08 regardless of the solution chosen.

Table 6 – Net Present Value of Alternative 1

Alternative 1	$i = 0.79\%$
0	\$(50,604.49)
1	\$225,776.08
2	\$225,776.08
3	\$225,776.08
4	\$225,776.08
5	\$225,776.08
6	\$225,776.08
7	\$225,776.08
8	\$225,776.08
9	\$225,776.08
10	\$225,776.08
11	\$225,776.08
12	\$225,776.08
NPV	\$2,504,466.27

Table 7 – Net Present Value of Alternative 2

Alternative 2	$i = 0.79\%$
0	\$(50,244.26)
1	\$225,776.08
2	\$225,776.08
3	\$225,776.08
4	\$225,776.08
5	\$225,776.08
6	\$225,776.08
7	\$225,776.08
8	\$225,776.08
9	\$225,776.08
10	\$225,776.08
11	\$225,776.08
12	\$225,776.08
NPV	\$2,504,823.67

The cashflows for economic yield were best modeled over a 1 year span. The rate used per month was 0.79%, which ultimately resulted in an alternative 1 net present value of \$2,504,466.27 and an alternative 2 net present value of \$2,504,823.67. In addition to net present value, an internal rate of return analysis (IRR) was performed to measure the comparative difference in return between both solutions. The alternative 1 IRR was 446%, and the alternative 2 IRR was 449%. While alternative 2 has a higher return on investment, it is important to consider the fact that alternative 1 has a greater ease of implementation to Rubbercraft's calendaring operation. The datalogging device produced by Mitutoyo connects directly to measuring equipment already used on the manufacturing floor, which means less change needs to occur. Alternative 2 is a change to the point of measurement, which involves more complexities in integration. Rubbercraft ultimately needs to make the decision on whether they prefer ease of manufacturing integration or higher return on their investment.

Table 8 – Internal Rate of Return Cashflows

Alternative 1 Cash Flows	
0	\$(50,604.49)
1	\$225,776.08
2	\$225,776.08
3	\$225,776.08
4	\$225,776.08
5	\$225,776.08
6	\$225,776.08

7	\$225,776.08
8	\$225,776.08
9	\$225,776.08
10	\$225,776.08
11	\$225,776.08
12	\$225,776.08
IRR	446%
Alternative 2 Cash Flows	
0	\$(50,244.26)
1	\$225,776.08
2	\$225,776.08
3	\$225,776.08
4	\$225,776.08
5	\$225,776.08
6	\$225,776.08
7	\$225,776.08
8	\$225,776.08
9	\$225,776.08
10	\$225,776.08
11	\$225,776.08
12	\$225,776.08
IRR	449%

Payback Period = 7 days

2.8. Monitoring and Process Control

The control chart system employs a dual-phase approach to address both immediate process adjustments and sustained quality improvement. For short-

term process control, operators generate I-MR charts on the 40" HD monitor display during production runs, plotting thickness measurements at consistent two-minute intervals (alternative 1) or 50 Hz sampling rate (alternative 2). These real-time charts display dynamic control limits calibrated to current material specifications. When measurements breach these limits, operators immediately redirect the affected rubber sheet through the calendaring process for corrective reprocessing. This rapid intervention protocol minimizes material waste while maintaining production continuity.

For long term quality oversight, quality engineers analyze aggregated control charts compiled from multiple production batches. These comprehensive charts employ statistically derived control limits based on the established process capability metrics ($C_p = 0.73$, $C_{pk} = 0.71$, per Table 1). Once a week, the quality engineering department focuses on identifying significant patterns that indicate systemic issues requiring intervention. Recurring trends across seven consecutive measurements prompt immediate machine calibration checks, while clustered measurements near control limits often reveal underlying mechanical wear. Cyclical variations in the data may correlate with material inconsistencies in the supplier's batches.

Upon detecting these patterns, engineers authorize targeted corrective measures ranging from preventive maintenance to system calibration. This tiered approach ensures operators maintain daily production quality while providing engineers with the diagnostic tools needed for equipment optimization. All chart data is systematically archived within the quality database Figure 10, creating an

auditable record for both continuous improvement initiatives and compliance verification.

2.9. Out-of-Control Action Plan

When out-of-control conditions are detected, operators follow the standardized Out-of-Control Action Plan (OCAP). This protocol begins with halting production to investigate potential root causes, including verification of sensor calibration and roller temperature settings. After implementing corrective actions, operators must confirm three consecutive in-spec measurements before resuming normal production. All incidents and adjustments are documented in the quality database for future analysis and process improvement.

The responsibility for maintaining process control is divided among team members, as illustrated in Figure 13. Operators perform thickness measurements and initiate the OCAP when needed. Quality technicians conduct random audits of 20% of production runs and maintain calibration records. Supervisors review control charts weekly and approve any permanent modifications to process parameters. Responsibilities and delegations are mapped out as a series of actions in the swim lane diagram.

Calendaring Machine: Out-of-Control Action Plan					
RUBBERCRAFT	Identify Issue	Containment	Root Cause Analysis	Corrective Action	Verify & Close
Operator	Detects non-conforming material	Stops production or flags affected parts	Provides input on machine settings and recent changes	Adjusts machine settings accordingly	Resumes production with corrected parameters
Quality Engineer	Reviews control chart data to confirm deviation		Reviews historical data and trends	Monitors control charts to ensure stability	Confirms process is back within specifications
Production Supervisor	Flags issues if specification limits are exceeded	Decides whether to continue, scrap, or rework material			Signs off on corrective actions and ensures documentation
Process Engineer			Conducts root cause analysis (Fishbone Diagram, 5 Whys)	Recommends process adjustments (roller pressure, temperature, speed)	
Control Chart	Flags issues if control limits are exceeded				

Figure 13 – Out of Control Action Plan

Process performance is evaluated through three key metrics: daily examination of control charts for immediate process stability, weekly analysis of Cp/Cpk values to assess long-term capability (with a target $C_p \geq 1.33$), and monthly tracking of scrap rates with the goal of reducing defects from the current 2.72% to below 1.5%. This multi-layered approach provides both real-time control and strategic process improvement.

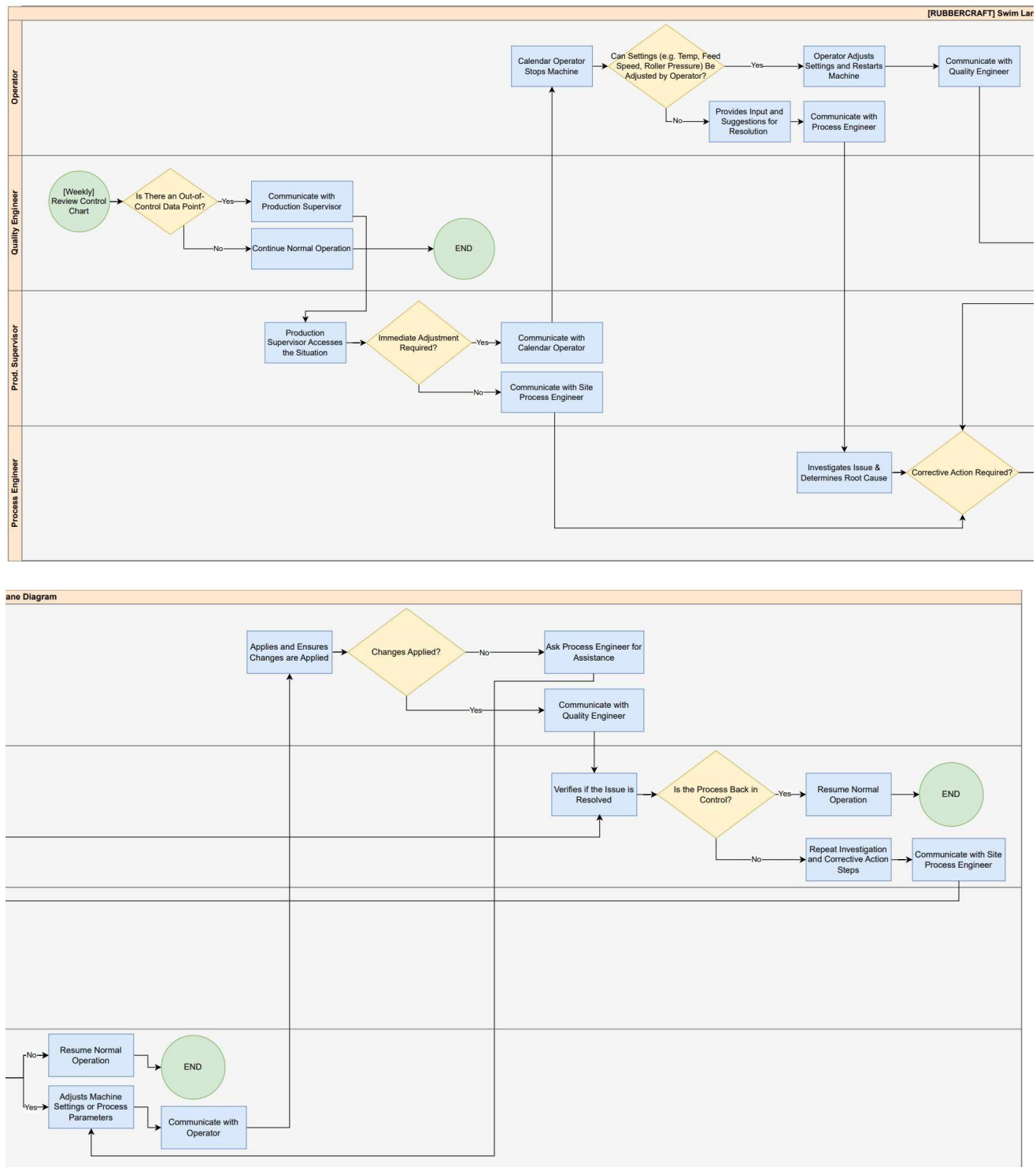


Figure 14 – Swim Lane Diagram, Halved for Viewability; Full Diagram in End Section of Report

2.10. Additional Recommendations

Beyond the immediate implementation of the proposed quality control system, there are several improvements that could further improve the calendaring operations at Rubbercraft.

2.10.1. Predictive Maintenance Program

The program would address the thickness variability caused by mechanical wear, uncalibrated rollers, and any issues that affect the quality of the rubber sheets exiting the calendar. Through analysis of historical data and potential data trends, maintenance intervals would be dynamically adjusted based on past measurements and upcoming setup. Rather than the typical preventative maintenance that occurs at set intervals, the program would function with the next production in mind. Variables such as the material, operators, and last maintenance date could be used to determine if maintenance is needed prior to production of the next roll using a linear programming model.

2.10.2. Workforce Development

To increase operator proficiency, a formalized training curriculum can help accelerate the adjustment to the calendaring process. This program would use hands-on thickness gauge practice with mentorship from experienced operators. Cross training with operators from other departments could help improve other processes given their suggestions with implementing changes to other machinery.

2.10.3. System Enhancements

The Arduino-based measurement system could be expanded to encompass the span of either half or entire sheet rather than just the edge. Given that the edge is typically the thickest part of the sheet produced during the calendaring process, it would be beneficial to ensure the thickness of the center (which is typically the thinnest).

To enhance precision and reduce operator dependency, Rubbercraft could retrofit the manual crank with a motorized digital adjustment system. This upgrade would allow operators to input target thickness values (e.g., 1.85 mm) via an industrial touchscreen, where servo motors automatically crank the calendar rolls to position. This is standard in most modern calendaring machines. The system could integrate with the quality database Figure 12 to store settings for repeat orders and have a programmable logic controller to actively adjust during production. This would require motors, encoders, and a control interface. Our team believes this retrofit could improve consistency without the need for a full machine replacement.

2.10.4. Continuous Improvement Process

Instituting monthly quality review meetings would formalize improvement efforts. These cross-functional sessions would analyze control chart trends, evaluate scrap rates by product type, and prioritize projects based

on ROI. Participation should include representatives from production, quality assurance, engineering teams, and shop-floor operators.

2.11. Standards and Constraints

As mentioned in the “Project Approach” section, this project followed the IISE industry standard of process improvement known as DMAIC. This is a process and quality improvement project, which is at the heart of Industrial and Manufacturing Engineering curricula. The DMAIC methodology was used to model every action taken from initial problem definition to implementation and control. The project holds true to industry standards to ensure solutions are adequate and meet quality requirements. The project integrates ISO 9001 standards to ensure that quality control measures are set within the calendaring process. Quality management techniques, like Statistical Process Control (SPC) and root cause analysis, are at the heart of our proposed solution to monitor thickness consistency and enhance overall quality.

Constraints are often present in all process improvement projects, especially in manufacturing. The management of these constraints ultimately determines whether a project is successful or not. Impediments to the completion of this project included budget constraints, time constraints, technological constraints, and resource constraints.

Budget constraints were present in the form of limitations in the amount of money the company was willing to spend on their solution. Initially the idea of replacing the entire calendaring machine due to its age and difficulty of operation was posed, however, the company would have to spend upwards of \$13,000 just

to purchase a new machine. This was an option they simply could not afford, so the solutions drafted needed to be much more cost effective while still yielding a comparable level of defect mitigation.

Time constraints were also present in the form of completing the project within a span of 8 months. This in combination with contrasting schedules, class and work commitments, along with company availability made collecting data a challenge. With this in mind our proposed solution would account for quick and effective material thickness monitoring, measuring, and recording of such data. This would make it very simple to continue long term. In terms of resource constraints, only one team member was able to reliably access valuable company data given their status within the company. These constraints were ultimately overcome through effective communication between team members and the company. Communication skills are often a key component to problem solving that is too often overlooked in problem solving, so the team's ability to utilize their skills in this area were essential to project success.

Technological constraints arose in the form of limited access to technology. The team had to rely on Microsoft excel and access to process data. There are more powerful software being commonly used in the field today, however, the team was able to successfully use their problem-solving prowess and educational experience to make these software work for their intended roles in solution construction. In order to combat these constraints, the team focused on delivering solutions with cost effectiveness in mind. Proposing alternatives for the company

to choose what they believe is best. Whether they prefer ease of implementation or do they prefer a higher rate of return.

3.0 Reflective Writing

3.1. Introduction

Our Senior Project was the first introduction for many of our team members to exploring the realm of real-world manufacturing problem solving. This project involved collaboration with two companies, Mitutoyo America and Rubbercraft Corporation of California, in order to implement a meaningful process and quality improvement to Rubbercraft's material calendaring operation. Our objective was to measure samples of rubber sheets exiting the calendaring rolls at Rubbercraft. This was done to analyze the magnitude of process variation actively occurring within production. Once our analysis was completed, we developed two solutions. One involved collaboration with a manufacturer of quality control products Mitutoyo America in order to use their data logging technology. This would allow us to easily transfer data from their measurement gages straight to a database where we could store and transform the data into a system that would allow engineers and operators to actively monitor the status of the operation. The second solution alternative involved an alteration to their point of measurement, which utilized a Time-of-Flight sensor fixed to the calendaring machine that would accurately measure the material and send the data to a database for the aforementioned purpose. An out-of-control action plan was also developed to ensure that the alternatives could be maintained and controlled once the project is out of our hands. We drafted multiple models that prove the economic feasibility

of both of these solution alternatives, and the preferred alternative is set to be used and implemented by Rubbercraft in the near future. We learned a lot about applying our educational skills to real life manufacturing and the obstacles that accompany a project like this. This paper illustrates a reflection on some of the obstacles and outcomes we internalized for our future careers.

3.2. Body

We took public health and safety into consideration with our circuit design by enclosing the device in a fixture that would shield operating personnel from potentially dangerous electrical components. We also made sure the fixture was drafted to be securely fastened to the calendaring machine to avoid it potentially becoming dislodged and pose a danger to operating personnel by being in close proximity to metal objects that are in constant motion. We did this using CAD software, and also created a potential development plan for Rubbercraft to execute within their prototyping department so that the fixture will feasibly withstand long-term implementation in their manufacturing environment.

Careers. There were a variety of concepts we had to learn on our own that were not necessarily covered in our degree program curriculum. We performed in-depth research on calendaring processes. We learned how to apply database design principles to manufacturing settings. When initially completing our Systems Engineering course, we only had experience applying database design to service settings like banks and rideshare services. Manufacturing databases often are much more detailed and need to conform to datatypes that are present in manufacturing operations and also require an engineering understanding of the

datatypes organized in order to accurately construct the database. We also learned how to create an out-of-control action plan. This is a concept that is not covered by our quality control class, or any other courses related to operations and manufacturing control. Having an out-of-control action plan is essential to proper maintenance of a manufacturing operation, and it is widely used in all manufacturing industries across America. We took it upon ourselves to learn how to do this as it applied to our project.

3.3. Conclusion

Engineering design heavily relies upon constraints unique to each manufacturer. Rubbercraft had their own set of constraints that can only be found in the manufacturing of aerospace grade polymers, and we had to cater our solutions to these constraints in order to work with them effectively. The absence of consideration of these constraints will not result in a usable product for the company. This project highlighted the importance of integrating quality and continuous improvement into technical real world problem-solving. Through our focus on the calendaring process, we gained firsthand experience in applying engineering design principles to address real manufacturing challenges. The process helped us to broaden our understanding of precision manufacturing and its impact on quality of material that affect performance and customer satisfaction. Moving forward, we recognize the value of approaching projects with a systems and quality mindset.

4.0 Conclusion and Recommendations

The main finding of the project reveals the current calendaring process has a 2.7% probability of producing a defect in CTQ thickness. It is important to identify, document, and continually monitor the defect because it enables corrective actions to occur in a timely manner. Addressing issues at their source and not during the final stages of inspection before parts are delivered to customers. The outcome of this project shines a light on the process in order to monitor the quality of materials produced in the process of calendaring and provides an effective system to continually measure, record, and analyze data. Material thickness was the one CTQ the project aimed to monitor and control. The process displays other types of defects including foreign object debris (FOD), voids, impressions, and air bubbles. These defects may be the result of the types of ingredients used in the materials. The number of reworks is also of concern. Why is material not being delivered to the next step in production correctly the first time? These are the questions further studies should aim to address. An initial assessment should be done to account for the time lost to materials reworked due to voids, impressions, air bubbles, and FOD.

One application of this project is the use of databases to track specified engineering characteristics which can be applied across different product types when the target value and measured values are known. This method would showcase who, when, what, and how often parts deviate from the specified requirements. One location to measure, monitor, and evaluate trends over time efficiently.

It is significant to study the quality of material being produced for final production parts because it can directly impact the performance, reliability, safety, and cost effectiveness

of the products delivered to the customer. In order to uphold the company's commitment to quality and serving its customer meeting on time delivery, monitoring the material allows to maintain control over the production of parts and identify opportunities for further improvement. The company services industries that must adhere to strict quality standards. Ensuring compliance with standards set by the industry is incredibly important for continual success in the business. Process control, production efficiency, and customer satisfaction all play a key role in establish brand reputation. As a result, customers gain confidence in knowing which supplier they can trust for their parts needs.

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6.0 Appendix

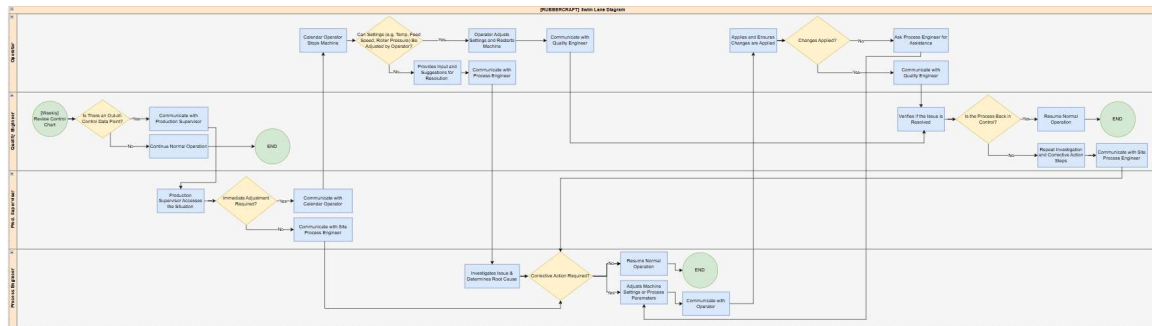


Figure 15 – Full View of the Swim Lane Diagram

PARTNUMBERTable

PARTNUMBERID

PART NUMBER

Figure 16 Data Entry Part Number Form

ORDERTable

ORDERID

QTY

CUSTOMER PART #

PART DESCRIPTION

CUSTOMERID

Figure 17 Data Entry Order Form

MFGTable

MFGID

MFG#

COMPOUND#

BATCH#

FABRIC PO#

ADHESIVE#

SEQ#

DEPARTMENT

CODE

DATE

OPERATOR NAME

TARGET

TOLERANCE

PART NUMBER

Figure 18 Data Entry Manufacturing Order Form

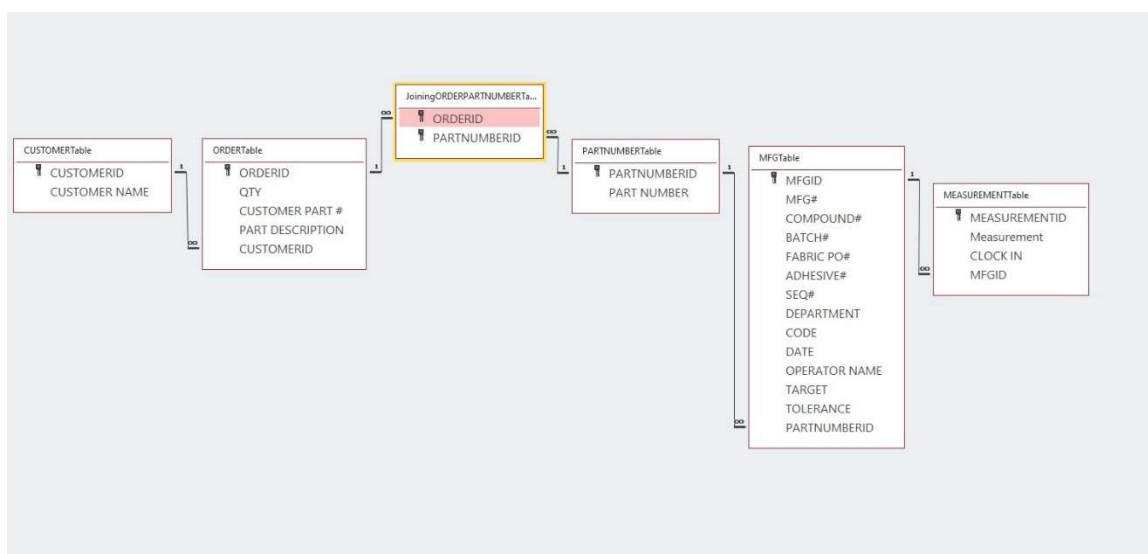
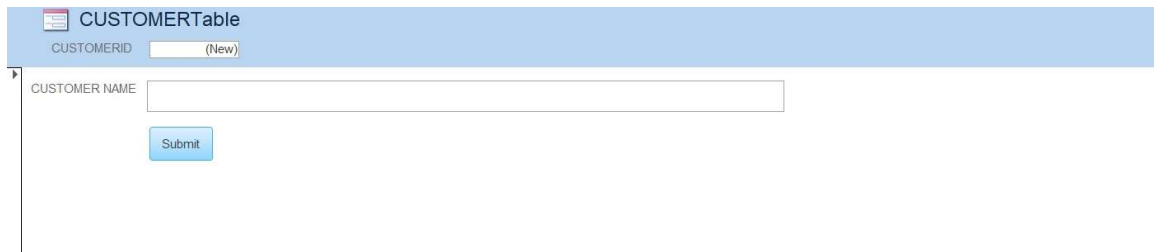


Figure 19 Data Relationships



CUSTOMERTable

CUSTOMERID (New)

CUSTOMER NAME

Figure 20 Data Entry Customer Form

```

DualSensorProgramV2Official.ino
1  /* Originally from https://forum.arduino.cc/t/using-2-vl53l0x-tof-sensors-with-arduino-uno/1312965. Credit goes to user "4nodews" for the base code.
2  Modified by Jordan Sycip, March 2025
3  Modifications:
4  - Displays "Thickness" by taking both sensor readouts and subtracting them from a set distance
5  - Tare button, requires pull-up resistor, to change the set distance to their current distance (e.g. sensors are positioned closer)
6  Instructions:
7  - Run the program on your Arduino and in the serial monitor set the Hertz (Hz) to match the serial begin as 9600
8  - To get measurement in inches, divide the readout by 25.4
9  */
10 #include <Wire.h>
11 #include <VL53L0X.h>
12
13 // Define VL53L0X sensor objects
14 VL53L0X sensor1;
15 VL53L0X sensor2;
16
17 // Define pins
18 #define XSHUT_1 2 // Shutdown pin for Sensor 1
19 #define XSHUT_2 3 // Shutdown pin for Sensor 2
20 int buttonStatus = 0;
21 int setDistance = 0;
22
23 void setup() {
24   Serial.begin(9600); // Initialize serial monitor
25   Wire.begin(); // Start I2C communication
26
27   // Set XSHUT pins as outputs
28   pinMode(XSHUT_1, OUTPUT);
29   pinMode(XSHUT_2, OUTPUT);
30
31   // Ensure both sensors are turned off before setting addresses
32   digitalWrite(XSHUT_1, LOW);
33   digitalWrite(XSHUT_2, LOW);
34   delay(50);
35
36   // Turn on Sensor 1 and initialize
37   digitalWrite(XSHUT_1, HIGH);
38   delay(50);
39   if (!sensor1.init()) {
40     Serial.println("Error: Sensor 1 failed to initialize!");
41     while (1); // Halt execution if initialization fails
42   }
43   sensor1.setAddress(0x30); // Assign a new I2C address
44   Serial.println("Sensor 1 initialized at address 0x30");
45
46   // Turn on Sensor 2 and initialize
47   digitalWrite(XSHUT_2, HIGH);

```

Figure 21 Material Thickness Algorithm Section 1

```

DualSensorProgramV2Official.ino
48   delay(50);
49   if (!sensor2.init()) {
50       Serial.println("Error: Sensor 2 failed to initialize!");
51       while (1);
52   }
53   sensor2.setAddress(0x31); // Assign a different I2C address
54   Serial.println("Sensor 2 initialized at address 0x31");
55
56   // Set timeout to prevent hanging
57   sensor1.setTimeout(500);
58   sensor2.setTimeout(500);
59
60   // Start continuous readings
61   sensor1.startContinuous();
62   sensor2.startContinuous();
63
64   // Tare button setup
65   pinMode(4, INPUT_PULLUP);
66   Serial.begin(9600);
67 }
68
69 void loop() {
70     uint16_t distance1 = sensor1.readRangeContinuousMillimeters();
71     uint16_t distance2 = sensor2.readRangeContinuousMillimeters();
72     {
73         Serial.print("Material Thickness: ");
74         Serial.print(setDistance-(distance1+distance2));
75         Serial.println(" mm");
76         // Check for timeout on Sensor 1
77         if (sensor1.timeoutOccurred()) {
78             Serial.println("Sensor 1 timeout!");
79         }
80
81         // Check for timeout on Sensor 2
82         if (sensor2.timeoutOccurred()) {
83             Serial.println("Sensor 2 timeout!");
84         }
85         delay(500); // Short delay between readings
86     }
87     int pinValue = digitalRead(4);
88     delay(10);
89     if (buttonStatus != pinValue) {
90         buttonStatus = pinValue;
91         setDistance = distance1+distance2;
92         Serial.println("Tare");
93     }
94 }

```

Figure 22 Material Thickness Algorithm Section 2